

WQBEL-TOXICS RECOMMENDATIONS

FACT/DECISION SHEET

Cadillac WWTP- (MI0020257- Reissuance)

PREPARED BY: Glen Schmitt

Today's Date: 5/2/2013

Background

The City of Cadillac submitted an NPDES permit reissuance application in March 2012 for the Cadillac WWTP. The facility is located in Cadillac, Michigan, Wexford County. The current NPDES permit was issued on September 5, 2008, effective on November 1, 2008, and is scheduled to expire on October 1, 2012. The current NPDES permit authorizes to continuously discharge a maximum of 3.2 million gallons per day (MGD) of treated municipal wastewater through outfall 001 to the Clam River. The design flow of 3.2 MGD is not to be considered a limitation or actual capacity. The Clam River upstream of outfall 001 has a 95 percent, harmonic mean, and 90dQ10 low flows of 2.7, 13, and 3.3 cubic feet per second (cfs), respectively (Low Flow File No. 8067). A hardness value of 130 mg/l CaCO₃ was used to develop Water Quality Based Effluent Limits (WQBELs) for parameters that are hardness dependent. (Source: average of hardness collected during January 2011 CSI (96 mg/l) and September 2009 CSI (162 mg/l) at Cadillac WWTP, samples collected at Clam River upstream of outfall 001). The Clam River is not protected as a drinking water source. The Clam River downstream of the proposed discharge from the Wexford/Missaukee County Line downstream to Turnerville Road is classified as a Blue Ribbon Trout Stream. The outfall 001 discharges to a reach that is not designated as a coldwater stream.

The NPDES permit application includes information about several Industrial Users discharging to the Cadillac WWTP.

Sources

Sources of information used for this review to develop the water quality-based effluent limits (WQBELs) include the facility's NPDES permit reissuance application, the Permits Section Toxics facility file, the current NPDES permit, discharge monitoring data, Compliance Sampling Inspections (CSIs) conducted in January 2011 and September 2009, and United States Geological Survey topographical maps. Based upon our review of this information, we have the following recommendations:

Recommendations

Outfall 001A:

- 1) The 2012 Water Quality and Pollution Control in Michigan Sections 303(d), 305(b), and 314 Integrated Report indicates the Clam River is Not Supporting Water Quality Standards (Category 5 Waterbody) for the Fish Consumption designated use due to Polychlorinated biphenyls (PCBs). A Total Maximum Daily Load for PCBs is scheduled to be developed in 2013. The nonattainment issues for this receiving water listed in the Integrated Report are not due to a pollutant that pertains to this discharge.
- 2) The current NPDES permit includes daily minimum and maximum pH limits of 6.5 and 9.0 Standard Units (S.U.), respectively with samples collected 5X/week as a grab sample. Levels of pH in the outfall 001 final effluent since January 2008 ranged from 5.5 to 8.7 S.U., averaging 7.3 S.U. There were periodic exceedances of the daily minimum pH final effluent limitation. Based on this information, we recommend the draft permit retain the current limits and monitoring requirements for pH.
 - a. These limits will allow the receiving water body to maintain the hydrogen ion concentration according to Rule 323.1053 of the Part 4 Michigan Water Quality Standards (WQS).
- 3) The NPDES permit reissuance application indicates ultraviolet disinfection is utilized to achieve compliance with the fecal coliform limitations instead of chlorine. Based on this information, we have no recommendations for total residual chlorine at this time.
- 4) The current NPDES permit includes a monthly average total phosphorus final effluent limit of 0.5 milligrams per liter (mg/l) (13 lbs/day) from May 1 through March 31 and a monthly average total phosphorus final effluent limit of 0.65 mg/l (17 lbs/day) from April 1 through April 30. Compliance with these limitations is 5X/week with samples collected as 24-hour composite samples. Concentrations of total phosphorus in the outfall 001 final effluent since January 2008 ranged from 0.0 to 1.78 mg/l (0.0 to 10.39 lbs/day). To protect the Clam River from excessive nutrients, we recommend the draft permit retain the current limitations and monitoring requirements for total phosphorus.
 - a. These final effluent limitations will protect the receiving water body and downstream waterbodies from excessive nutrients and is consistent with Rule 323.1060 of the Part 4 Michigan WQS.
 - i. Historically, the Clam River was known for excessive plant growth due to excessive nutrients from point source discharges, which is why the Cadillac WWTP has specific total phosphorus limitations. These limitations were included several permit cycles ago (probably late 80's or early 90's) due to excessive nutrient issues in the Clam River from the Cadillac WWTP. During the 1970s and 1980s, issues with excessive nutrients were documented with the primary source being the Cadillac WWTP.
 - ii. The last few permit cycles kept the current limitations for total phosphorus. No nuisance plant growth was observed downstream of the outfall in recent surveys. While the concentration limitations have been the same for several permit cycles, the loading limitations may have increased when the discharge volume increased.
 - b. The Haring Twp WWTP (MI0059076) is proposing to construct a new WWTP to service the township. The facility has yet to be constructed, however the proposed discharge location is approximately 1.5 miles downstream of the current outfall location for the Cadillac WWTP. Haring Township is currently sending their effluent for treatment at the Cadillac WWTP.
 - c. Any increase of design flow or a new use should not allow additional loading of nutrients to the Clam River
 - d. The four current NPDES permitted point sources are permitted to discharge a combined flow of 8.98 MGD (13.9 cfs). The Cadillac WWTP includes final effluent limitations for total phosphorus and discharges a maximum volume of 3.2 MGD (4.95 cfs). The Clam River near the proposed outfall from the Haring Twp WWTP is effluent dominated, with the final effluent from the Cadillac WWTP comprising approximately 50% of the lowest 95 percent exceedance low flow of the Clam River.
- 5) The current NPDES permit includes a level currently achievable total mercury limit of 10 nanograms per liter (ng/l) [0.00027 lbs/day] with compliance determined quarterly as a 12-month rolling average. A Pollutant Minimization Program for total mercury was approved on May 17, 2005. Concentrations of total mercury in the outfall 001 final effluent since July 2008 ranged from Not Detected to 4.7 nanograms per liter (ng/l). These data indicate there is a reasonable potential for total mercury to be discharged at concentrations exceeding Michigan Water Quality Standards (WQS) in the outfall 001 final effluent. We recommend the draft permit include a level currently achievable total mercury final effluent limit of 2.0 ng/l (0.00005 lbs/day). Compliance with these limitations is calculated as a 12-month rolling average with samples collected quarterly as grab samples. The PMP for total mercury approved in May 2005 should be retained in the draft NPDES permit.
 - a. The PMP annual updates for total mercury were not available in NMS. The PMP was approved in May 2005.

- b. The final effluent data for total mercury appears to be in compliance with the 12-month rolling average final effluent limitations.
 - c. Review of total mercury final effluent data and reasonable potential analysis was consistent with Water Resources Division, Policy and Procedure, WRD-004, Calculation of Level Currently Achievable (LCA) for Mercury in Proposed National Pollutant Discharge Elimination System (NPDES) Permits.
- 6) The current NPDES permit includes a quarterly monitoring requirement for total selenium with samples collected as 24-hour composite samples. Concentrations of total selenium in the outfall 001 final effluent since November 2008 ranged from Not Detected to 5 micrograms per liter (ug/l). Based on this data, there is not a reasonable potential for total selenium to be discharged at concentrations exceeding Michigan WQS. We recommend the draft permit remove the quarterly monitoring requirement for total selenium.
- 7) The current NPDES permit includes a quarterly monitoring requirement for available cyanide with samples collected as grab samples. Concentrations of available cyanide in the outfall 001 final effluent since November 2008 ranged from Not Detected to 17 ug/l. Based on this data, there is a reasonable potential for cyanide to be discharged at concentrations exceeding Michigan WQS. We recommend the draft permit include a monthly average WQBEL for available cyanide of 5.9 ug/l (0.16 lbs/day). Compliance with these limitations should be weekly with samples collected as grab samples and analyzed using EPA Method OIA-1677 with a quantification level of 2 ug/l. Appropriate language should also be included in the draft permit to allow the opportunity for a monitoring frequency reduction from weekly to monthly after one year.
- 8) The current NPDES permit includes a quarterly monitoring requirement for total zinc with samples collected as grab samples. Concentrations of total zinc in the outfall 001 final effluent since August 2008 ranged from Not Detected to 715 ug/l. Based on this data, there is not a reasonable potential for total zinc to be discharged at concentrations exceeding Michigan WQS. We recommend the draft permit remove the quarterly monitoring requirement for total zinc.
- 9) Concentrations of total copper in the outfall 001 final effluent since August 2009 ranged from 4.0 to 15 ug/l. Based on this data, there is a reasonable potential for total copper to be discharged at concentrations exceeding Michigan WQS. We recommend the draft permit include a monthly average WQBEL for total copper of 19 ug/l (0.5 lbs/day). Compliance with these limitations should be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA method. Appropriate language should also be included in the draft permit to allow the permittee to request a monitoring frequency reduction from weekly to monthly after one year.
- 10) The current NPDES permit includes final effluent limitations for chronic whole effluent toxicity (WET) of 1.1 chronic toxic units (TUC). Compliance with these limitations are monthly with samples collected as 24-hour composite samples and a reporting requirement for acute toxicity. Required test species is *Ceriodaphnia dubia* (*C. dubia*). The outfall 001 final effluent was never acutely toxic to *C. dubia* (all tests reported as 0.0 acute toxic units (TUa)). Chronic toxicity ranged from 0.0 to 2.4 TUC. These data indicate the outfall 001 final effluent does not have the reasonable potential to be acutely toxic. However, these data indicate the outfall 001 final effluent does have the reasonable potential to be chronically toxic. Based on this information, we recommend the draft permit retain the current limits and monitoring requirements for WET with the exception the monitoring frequency be reduced from monthly to quarterly. In addition, fathead minnow WET testing should be required under the conditions found in Part I.A.2, Additional Monitoring Requirements of the current NPDES permit. Four fathead minnow WET tests should be conducted consistent with the dates specified in the reissued permit for the Additional Monitoring Requirements.
- a. The permittee was approved to only conduct toxicity testing on *C. dubia* for compliance with the current permit final effluent limitations.
 - b. The monitoring frequency can be reduced from monthly to quarterly as no WET tests were acutely toxic, only 2 of 57 WET tests were chronically toxic, and the level of chronic toxicity was low, but above the current limitations.
 - c. Fathead minnow testing should be required under the Additional Monitoring Requirements. Federal Regulations require testing under these requirements using two species.
- 11) All other data reviewed indicated there was not a reasonable potential for these parameters to be discharged at concentrations exceeding Michigan WQS. We have no recommendations for these parameters at this time. Review PEL sheet for list of other parameters reviewed.

NPDES PERMIT REVIEW SUMMARY

Facility:	Cadillac WWTP
Permit Number:	MI0020257
Reviewer:	Glen Schmitt - Lakes Michigan and Superior Permits Unit, Permits Section, WRD, MDEQ
Date:	5/2/2013

Outf.	Authorized Flow		Disch. Type *	Receiving Water	Lake?	FLOWS (cfs)			pH	Hardness
	mgd	cfs				95% Exc.	H. Mean	90Q10		
001	3.2	4.95	Municipal	Clam River	☐	2.7	13.0	3.3	7	130

* P = Process; N = Noncontact Cooling; C = Contact Cooling; S = Sanitary; ST = Storm Water; GW = Groundwater Purge

SOURCES REVIEWED

X **Permit Application**

Type of issuance: REISSUANCE

X **Facility/VGW File**

*Low Flows from Low Flow File No. 8067

X **NMS Documents & Files**

*Hardness value is average of values measured during Sept 2009 and Jan 2011 CSI's (162 & 96 mg/l)

X **Current Permit**

Parameters Limited:

Parameter	Mo Avg.	Load	D Max

Special Conditions:

WTA

WTA/NMS GUIDANCE

X **Biosurvey Reports**

Report #:

X **WET Tests**

Date:

X **MiSWIMS/STORET**



REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 1 of 2)

*concentration values in ug/L except as noted

Facility:	Cadillac WWTP	Hardness:	130 mg/L CaCO3
Outfall:	001	pH:	7
Date:	5/2/2013	Drinking Water:	n

Parameter	CAS #	Water Quality Values										Diss. Met. Translator	Background Conc.
		FCV	vd	HNW	vd	HCV	vd	WV	vd	FAV	vd		
Cyanide, free	57125	5.2	1199707	48000	1199707	NA	0	NA	0	44	1199707	1	0
Selenium	7782492	5	1199707	2700	1199704	NA	0	NA	0	120	1199808	1	0
Zinc	7440666	147.5493971	1199707	16000	1200510	NA	0	NA	0	292.7044836	1199707	2.1	13
Mercury @	7439976	0.77	1199707	0.0018	1199707	NA	0	0.0013	1199707	2.8	1199707	1	**
Arsenic #	7440382	150	1199707	280	1201009	10	1201009	NA	0	680	1199707	1	1
Barium	7440393	578.5947781	2200905	160000	1199705	NA	0	NA	0	3301.7317	2200905	1	0
Boron	7440428	7200	1201112	330000	1201209	NA	0	NA	0	69000	1201112	1	0
Chromium	7440473	91.88040863	1199707	9400	1199706	NA	0	NA	0	1412.68133	1199707	1.5	1.6
Copper	7440508	11.20641288	1199707	38000	1200512	NA	0	NA	0	34.41581322	1199707	1.5	4
Nickel	7440020	64.93129014	1199707	210000	1199706	NA	0	NA	0	1169.205016	1199707	1.1	3
#N/A	1	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A

= Carcinogen

** = Background exceeds standards.

REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 2 of 2)

**concentration values in ug/L except as noted; loads in lb/d*

Facility:	Cadillac WWTP	Disch. Rate:	4.95 cfs	95% Ex. Flow:	2.7 cfs	Conc.	Load (lbs/day)
Outfall:	001			H. Mean Flow:	13 cfs	Hg Loading Calculated based on LCA (ng/L) of :	2 0.00053376
Date:	5/2/2013			90Q10 Flow:	3.3 cfs	TP Loading based on concentration (mg/L) of :	0

Parameter	Monthly Average PEL								Daily Max PEL		PEQ		DECISION	
	FCV	load	HNV	load	HCV	load	WV	load	conc	load	Avg	Max	Avg	Max
Cyanide, free	5.90858125	0.157688	79492.5	2121.496	#VALUE!	#VALUE!	#VALUE!	#VALUE!	44	1.174272	8.4	8.4	Reasonable Potential. Recommend a Monthly Avg WQBEL of 5.9 ug/l (0.16 lbs/day), weekly monitoring.	
Selenium	5.681328125	0.151623	4471.45313	119.3341	#VALUE!	#VALUE!	#VALUE!	#VALUE!	120	3.20256	4.5	4.5	No Reasonable Potential. Remove monitoring requirements for total selenium.	
Zinc	350.3046936	9.348932	26488.9708	706.9377	#VALUE!	#VALUE!	#VALUE!	#VALUE!	614.6794	16.40456	43.5	643.5	No Reasonable Potential. Remove monitoring requirements for total zinc.	
Mercury @	0.77	0.02055	0.0018	4.8E-05	NA	#VALUE!	0.0013	3.4694E-05	PEQ >= +10 Samples	2.8	0.074726	46.88604	106.3461	Reasonable Potential. Recommend and LCA of 2.0 ng/l (0.00005 lbs/day), quarterly monitoring, and PMP.
Arsenic #	170.3035781	4.545062	463.050156	12.35788	15.90484	0.424468	#VALUE!	#VALUE!	PEQ >= +10 Samples	2.8	0.074726	0.00497	0.00497	
Barium	657.4373571	17.54569	264975	7071.653	#VALUE!	#VALUE!	#VALUE!	#VALUE!	680	18.14784	5.2	5.2	No Reasonable Potential. No Recommendations.	
Boron	8181.1125	218.3375	546510.938	14585.28	#VALUE!	#VALUE!	#VALUE!	#VALUE!	3301.732	88.11662	52	52	No Reasonable Potential. No Recommendations.	
Chromium	156.3827999	4.173544	15566.2315	415.4316	#VALUE!	#VALUE!	#VALUE!	#VALUE!	69000	1841.472	923	923	No Reasonable Potential. No Recommendations.	
Copper	18.55513011	0.495199	62928.9381	1679.448	#VALUE!	#VALUE!	#VALUE!	#VALUE!	2119.022	56.55246	0	0	No Reasonable Potential. No Recommendations.	
Nickel	80.7483154	2.155011	347777.719	9281.492	#VALUE!	#VALUE!	#VALUE!	#VALUE!	51.62372	1.377734	39	39	Reasonable Potential. Recommend a Monthly Avg WQBEL of 19 ug/l (0.5 lbs/day), weekly monitoring.	
									1286.126	34.32412	15	15	No Reasonable Potential. No Recommendations.	
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		

* SILVER - Appropriate Dissolved Metal Translator must be entered in PEL 1

PEQ -- Effluent Data Summary
(all values in ug/L)

DATA ENTRY																
Parameter:	Cyanide, free	Source	Selenium	Source	Zinc	Source	Mercury @	Source	Arsenic #	Source	Barium	Source	Boron	Source	Chromium	Source
No. Nondetect	16 DMRs		75 DMRs		69 DMRs		8 DMRs		1 Application		0 Application		0 Application		4 Application	
Data	4	11/24/2008	2	2/2/2012	54	8/31/2007	0.00195	7/31/2007	2	8/12/2009	16	8/12/2009	355	8/12/2009		
	5	5/6/2009	2	11/7/2012	55	2/29/2008	0.00175	8/31/2007	2	3/1/2011	18	5/4/2010	220	5/4/2010		
	6	1/24/2011 (CSI)	5	11/1/2008	715	11/24/2008	0.00055	9/30/2007	1	10/3/2011	15	3/1/2011	182	3/1/2011		
			5	5/1/2009	17.8	8/12/2009	0.0014	10/31/2007			20	10/3/2011	216	10/3/2011		
					18	9/8/2009 (CSI)	0.00185	11/30/2001								
					63	2/3/2010	0.0023	12/31/2007								
	Not Detected		Not Detected		61	2/9/2010	0.00355	1/31/2008	Not Detected						Not Detected	
	2/2/2009		8/12/2009		59	2/13/2010	0.0027	2/29/2008	5/4/2010						5/4/2010	
	8/12/2009		9/8/2009 (CSI)		69	1/24/2011 (CSI)	0.0011	3/31/2008							3/1/2011	
	9/8/2009 (CSI)		11/12/2009		167	2/2/2011	0.00055	4/30/2008							10/3/2011	
	11/12/2009		2/3/2010		56	2/11/2011	0.00095	6/30/2008							8/12/2009	
	2/2/2010		2/9/2010		77	2/12/2011	0.00095	10/27/2008								
	5/4/2010		2/13/2010		99	11/8/2011	0.0008	11/24/2008								
	8/2/2010		2/17/2010				0.0005	2/9/2009								
	11/1/2010		2/25/2010				0.0009	5/7/2009								
	2/2/2011		5/4/2010				0.0047	8/12/2009								
	5/2/2011		8/1/2010		Not Detected		0.001	9/8/2009								
	8/1/2011		8/2/2010		11/30/2007		0.0005	11/12/2009								
	11/1/2011		8/3/2010		5/31/2008		0.0005	2/16/2010								
	5/1/2012		8/4/2010		8/31/2008		0.00075	5/4/2010								
	8/1/2012		11/3/2010		11/1/2008		0.00055	8/2/2010								
	11/1/2012		11/6/2010		11/4/2008		0.0005	11/9/2010								
	2/4/2013		11/9/2010		11/12/2008		0.0008	1/24/2011								
			11/17/2010		11/20/2008		0.0006	2/8/2011								
			11/23/2010		2/3/2009		0.0006	5/2/2011								
			1/24/2011 (CSI)		2/7/2009		0.00075	2/2/2012								
			2/2/2011		2/9/2009		0.0008	5/9/2012								
			2/9/2011		2/17/2009		0.00085	2/6/2013								
			2/10/2011		2/26/2009											
			2/13/2011		5/1/2009											
			2/21/2011		11/12/2009			Not Detected								
			5/4/2011		2/17/2010			5/31/2008								
			5/7/2011		2/25/2010			7/31/2008								
			5/10/2011		5/4/2010			8/31/2008								
			5/19/2011		8/1/2010			9/30/2008								
			5/26/2011		8/2/2010			8/10/2011								
			8/3/2011		8/3/2010			11/1/2011								
			8/6/2011		8/4/2010			8/7/2012								
			8/12/2011		11/3/2010			11/5/2012								
			8/16/2011		11/6/2010											
		8/24/2011		11/9/2010												
		8/30/2011		11/17/2010												
		11/2/2011		2/15/2011												
		11/5/2011		5/4/2011												
		11/8/2011		5/7/2011												
		11/14/2011		5/10/2011												
		11/20/2011		5/19/2011												

PEQ -- Effluent Data Summary
(all values in ug/L)

Parameter:	Cyanide, free	Source	Selenium	Source	Zinc	Source	Mercury @	Source	Arsenic #	Source	Barium	Source	Boron	Source	Chromium	Source
				11/30/2011		5/26/2011										
				2/8/2012		8/3/2011										
				2/14/2012		8/6/2011										
				2/22/2012		8/12/2011										
				2/25/2012		8/16/2011										
				2/28/2012		8/24/2011										
				5/5/2012		8/30/2011										
				5/8/2012		11/2/2011										
				5/16/2012		11/5/2011										
				5/20/2012		11/14/2011										
				5/24/2012		11/20/2011										
				5/29/2012		11/30/2011										
				8/8/2012		2/2/2012										
				8/15/2012		2/8/2012										
				8/18/2012		2/14/2012										
				8/23/2012		2/22/2012										
				8/28/2012		2/25/2012										
				11/14/2012		2/28/2012										
				11/17/2012		5/5/2012										
				11/20/2012		5/8/2012										
				11/28/2012		5/16/2012										
				11/4/2008		5/20/2012										
				11/12/2008		5/24/2012										
				11/20/2008		5/29/2012										
				11/24/2008		8/8/2012										
				2/3/2009		8/15/2012										
				2/7/2009		8/18/2012										
				2/8/2009		8/23/2012										
				2/16/2009		8/28/2012										
				2/25/2009		11/7/2012										
				2/6/2013		11/14/2012										
				2/14/2013		11/17/2012										
				2/16/2013		11/20/2012										
				2/18/2013		11/28/2012										
				2/26/2013		2/6/2013										
						2/14/2013										
						2/16/2013										
						2/18/2013										
						2/26/2013										

PEQ -- Effluent Data Summary
(all values in ug/L)

Parameter:	Copper	Source	Nickel	Source	#N/A	Source	#N/A	Source	#N/A	Source	#N/A	Source	#N/A	Source	#N/A	Source	#N/A
No. Nondetect	0	Application	0	Application	0		0		0		0		0		0		0
Data	4	8/12/2009	5	8/12/2009													
	8	5/4/2010	3	5/4/2010													
	15	3/1/2011	5	10/3/2011													
	8	10/3/2011															
				Not Detected													
				3/1/2011													

PROJECTED EFFLUENT QUALITY ANALYSIS (<10 Detectable):

* all values in ug/l

	Cyanide, free	Selenium	Zinc	Mercury @	Arsenic #	Barium	Boron	Chromium	Copper	Nickel	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
Sample Max.	6	5	715	0.00355	2	20	355	0	15	5	0	0	0	0	0	0	0	0	0
Total Samples	19	79	78	17	4	4	4	4	4	3	0	0	0	0	0	0	0	0	0
Multiplier	1.4	0.9	0.9	1.4	2.6	2.6	2.6	2.6	2.6	3	0	0	0	0	0	0	0	0	0
PEQ:	8.4	4.5	643.5	0.00497	5.2	52	923	0	39	15	0	0	0	0	0	0	0	0	0

PROJECTED EFFLUENT QUALITY ANALYSIS (>=10 Detectable):

PARAMETER:	Cyanide, free		Selenium		Zinc		Mercury @		Arsenic #		Barium		Boron
TOTAL NO. VALUES:	19		79		82		36		4		4		4
DETECTED:	3		4		13		28		3		4		4
NON-DETECTED:	16		75		69		8		1		0		0
d (% data<detect)	0.842105		0.949367		0.841463		0.222222		0.25		0		0
m (mean of data >detect)	5		3.5		116.2154		0.001239		1.6666667		17.25		243.25
std dev	1		1.732051		183.6926		0.001014		0.5773503		2.2173558		76.42589
n	1	30	1	30	1	30	1	4	1	30	1	30	1
d^n	0.842105	0.005768	0.949367	0.21039	0.841463	0.005637	0.222222	0.002439	0.25	8.674E-19	0	0	0
p	0.683333	0.94971	0.0125	0.936678	0.684615	0.949717	0.935714	0.949878	0.9333333	0.95	0.95	0.95	0.95
std normal dist fn (Z)	1.516513	2.445383	0.158611	2.349262	1.519186	2.445436	2.342826	2.446749	2.3272517	2.4477468	2.4477468	2.4477468	2.4477468
Z_p	0.47662	1.642405	-1.986737	1.527779	0.480226	1.642468	1.520067	1.644027	1.5013852	1.6452114	1.6452114	1.6452114	1.6452114
1+(s/m)^2	1.04	1.04	1.244898	1.244898	3.498366	3.498366	1.668882	1.668882	1.12	1.12	1.0165231	1.0165231	1.0987132
(sigma_d)^2	0.039221	0.039221	0.219054	0.219054	1.252296	1.252296	0.512154	0.512154	0.1133287	0.1133287	0.0163881	0.0163881	0.0941397
mu_d	1.589828	1.589828	1.143236	1.143236	4.129297	4.129297	-6.949297	-6.949297	0.4541613	0.4541613	2.8396181	2.8396181	5.4470199
(sigma_dn)^2	0.039221	0.164989	0.219054	0.343888	1.252296	0.526281	0.512154	0.249426	0.1133287	0.0163107	0.0163881	0.0005506	0.0941397
mu_dn	1.589828	-0.313099	1.143236	-1.666118	4.129297	2.656188	-6.949297	-7.066806	0.4541613	0.2149882	2.8396181	2.8475368	5.4470199
P_95 exponent	1.684218	0.354027	0.213381	-0.770199	4.666699	3.84772	-5.861462	-6.245736	0.9595925	0.4251037	3.0502316	2.8861422	5.9518068
P_95 (PEQ)	5.388238	1.424794	1.237856	0.462921	106.3461	46.88604	0.002847	0.001939	2.6106324	1.529749	21.120236	17.924029	384.44734
	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max

NOTES: For purposes of this summary, ^ represents an exponent or superscript while _ represents a subscript.
Variables are defined in Part 12, Rule 1225 (3) (a).

	Chromium		Copper		Nickel		#N/A		#N/A		#N/A		#N/A	
	4		4		3		0		0		0		0	
	0		4		3		0		0		0		0	
	4		0		0		0		0		0		0	
	1		0		0		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!	
	#DIV/0!		8.75		4.33333333		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!	
	#DIV/0!		4.57347424		1.15470054		#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!	
30	1	30	1	30	1	30	1	30	1	30	1	30	1	30
0	1	1	0	0	0	0	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
0.95	#DIV/0!	#DIV/0!	0.95	0.95	0.95	0.95	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
2.4477468	#DIV/0!	#DIV/0!	2.44774683	2.44774683	2.44774683	2.44774683	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
1.6452114	#DIV/0!	#DIV/0!	1.64521144	1.64521144	1.64521144	1.64521144	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
1.0987132	#DIV/0!	#DIV/0!	1.27319728	1.27319728	1.07100592	1.07100592	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
0.0941397	#DIV/0!	#DIV/0!	0.24153128	0.24153128	0.06859832	0.06859832	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
5.4470199	#DIV/0!	#DIV/0!	2.04828806	2.04828806	1.43203791	1.43203791	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
0.003285	#DIV/0!	#DIV/0!	0.24153128	0.00906536	0.06859832	0.00236407	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
5.4924472	#DIV/0!	#DIV/0!	2.04828806	2.16452102	1.43203791	1.46515504	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
5.5867429	#DIV/0!	#DIV/0!	2.85684091	2.3211652	1.86293985	1.54514797	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
266.865	#DIV/0!	#DIV/0!	17.4064514	10.187538	6.44264938	4.68866536	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
Average	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average

REASONABLE POTENTIAL ANALYSIS: WET (pg. 1 of 2)

Facility: Cadillac WWTP
Outfall: 001

Qe	Qr	RWC	PEL		PEQ*	
			TU _C	TU _A	TU _C	TU _A
4.953560372	0.675	88.01	1.1362656	1	2.40	0.00

* PEQ (enter data for most sensitive species only):

Chronic

Total n = 58 Quant n = 2 CV = 0.6
TU_C effluent = 2.4 MF = 1 PEQ TU_C = 2.40

Acute

Total n = 58 Quant n = 0 CV = 0.6
TU_A effluent = 0 MF = 1 PEQ TU_A = 0.00

Decision (WET limits/conditions):

WQBEL		Species	TI/RE	Monitoring	
Monthly Avg	Daily Max			Short term	Long term
1.1	1.0	C. dubia		Quarterly	

Basis of decision:

No Reasonable Potential for Acute Toxicity. Based on the data, there is a reasonable potential for the outfall 001 final effluent to be chronically toxic. Recommend retaining WET limits and monitoring requirements. Require fathead minnow as a test species under the conditions in Part I.A.2, Additional Monitoring Requirements.

WET Data Summary

Species 1: Ceriodaphnia dubia

Lab	Test Date	TU		Reference
		Acute	Chronic	
	8/31/2007	0.0	0.0	DMRs
	11/30/2007	0.0	0.0	DMRs
	2/29/2008	0.0	0.0	DMRs
	5/31/2008	0.0	0.0	DMRs
	8/31/2008	0.0	0.0	DMRs
	11/4/2008	0.0	0.0	DMRs
	12/11/2008	0.0	0.0	DMRs
	1/20/2009	0.0	0.0	DMRs
	2/17/2009	0.0	0.0	DMRs
	3/1/2009	0.0	0.0	DMRs
	4/9/2009	0.0	0.0	DMRs
	5/7/2009	0.0	0.0	DMRs
	6/4/2009	0.0	0.0	DMRs
	7/9/2009	0.0	0.0	DMRs
	8/6/2009	0.0	0.0	DMRs
	9/10/2009	0.0	0.0	DMRs
	10/8/2009	0.0	0.0	DMRs
	11/12/2009	0.0	0.0	DMRs
	12/3/2009	0.0	0.0	DMRs
	1/7/2010	0.0	0.0	DMRs
	2/4/2010	0.0	0.0	DMRs
	3/4/2010	0.0	0.0	DMRs
	4/8/2010	0.0	0.0	DMRs
	5/20/2010	0.0	0.0	DMRs
	6/3/2010	0.0	0.0	DMRs
	7/8/2010	0.0	0.0	DMRs
	8/3/2010	0.0	0.0	DMRs
	9/9/2010	0.0	0.0	DMRs
	10/7/2010	0.0	0.0	DMRs
	11/3/2010	0.0	0.0	DMRs
	12/9/2010	0.0	0.0	DMRs
	1/6/2011	0.0	0.0	DMRs
	2/8/2011	0.0	0.0	DMRs
Paragon	3/3-3/10/2011	0.0	0.0	DMRs
	4/7/2011	0.0	0.0	DMRs
	5/5/2011	0.0	0.0	DMRs
	6/9/2011	0.0	0.0	DMRs
	7/12/2011	0.0	0.0	DMRs
	8/4/2011	0.0	0.0	DMRs
	9/8/2011	0.0	0.0	DMRs
Paragon	10/6-10/13/201	0.0	0.0	DMRs
	11/3/2011	0.0	0.0	DMRs
	12/8/2011	0.0	0.0	DMRs
	1/5/2012	0.0	0.0	DMRs
	2/9/2012	0.0	0.0	DMRs
	3/6/2012	0.0	<u>1.3</u>	DMRs
	4/5/2012	0.0	0.0	DMRs
	5/3/2012	0.0	0.0	DMRs
	6/7/2012	0.0	0.0	DMRs
	7/7/2012	0.0	0.0	DMRs
	8/9/2012	0.0	0.0	DMRs
	9/13/2012	0.0	<u>2.4</u>	DMRs
	10/4/2012	0.0	0.0	DMRs
	11/8/2012	0.0	0.0	DMRs
	12/15/2012	0.0	0.0	DMRs
	1/3/2013	0.0	0.0	DMRs
	2/6/2013	0.0	0.0	DMRs
	3/7/2013	0.0	0.0	DMRs

Modified 10/4/00